

3212. Count Submatrices With Equal Frequency of X and Y

Solved 

Medium

 Topics

 Companies

 Hint

Given a 2D character matrix `grid`, where `grid[i][j]` is either `'X'`, `'Y'`, or `'.'`, return the number of [submatrices](#) that contain:

- `grid[0][0]`
- an **equal** frequency of `'X'` and `'Y'`.
- **at least** one `'X'`.

Example 1:

Input: `grid = [ ["X", "Y", "."], ["Y", ".", "."] ]`

Output: 3

Explanation:

X	Y	.
Y	.	.

X	Y	.
Y	.	.

X	Y	.
Y	.	.

Example 2:

Input: grid = [["X","X"],["X","Y"]]

Output: 0

Explanation:

No submatrix has an equal frequency of 'X' and 'Y'.

Example 3:

Input: grid = [".","."], [".","."]]

Output: 0

Explanation:

No submatrix has at least one 'X'.

Constraints:

- $1 \leq \text{grid.length}, \text{grid}[i].\text{length} \leq 1000$
- $\text{grid}[i][j]$ is either 'X', 'Y', or '.'.

Python:

class Solution:

def numberOfSubmatrices(self, grid: List[List[str]]) -> int:

rows = len(grid)

cols = len(grid[0])

sumX = [0] * cols

sumY = [0] * cols

res = 0

for i in range(rows):

```

rx = 0
ry = 0
for j in range(cols):
    if grid[i][j] == 'X':
        rx += 1
    elif grid[i][j] == 'Y':
        ry += 1

    sumX[j] += rx
    sumY[j] += ry

    if sumX[j] > 0 and sumX[j] == sumY[j]:
        res += 1

return res

```

JavaScript:

```

const numberOfSubmatrices = grid => {
    const rows = grid.length;
    const cols = grid[0].length;
    const sumX = new Int32Array(cols);
    const sumY = new Int32Array(cols);
    let res = 0;

    for (let i = 0; i < rows; i++) {
        let rx = 0;
        let ry = 0;

        for (let j = 0; j < cols; j++) {
            if (grid[i][j] === 'X')
                rx++;
            else if (grid[i][j] === 'Y')
                ry++;

            sumX[j] += rx;
            sumY[j] += ry;

            if (sumX[j] > 0 && sumX[j] === sumY[j])
                res++;
        }
    }

    return res;
};

```

Java:

```
class Solution {
    public int numberOfSubmatrices(char[][] grid) {
        int rows = grid.length, cols = grid[0].length;
        int[] sumX = new int[cols];
        int[] sumY = new int[cols];
        int res = 0;

        for (int i = 0; i < rows; i++) {
            int rx = 0, ry = 0;

            for (int j = 0; j < cols; j++) {
                if (grid[i][j] == 'X') rx++;
                else if (grid[i][j] == 'Y') ry++;

                sumX[j] += rx;
                sumY[j] += ry;

                if (sumX[j] > 0 && sumX[j] == sumY[j]) res++;
            }
        }

        return res;
    }
}
```